

Cloudflare Radar

2024 Year in Review

A data-driven analysis of global Internet traffic, adoption trends, connectivity disruptions, and security threats observed through Cloudflare's global network during 2024.

17%

Global Traffic Growth

225

Major Disruptions

6.5%

Traffic Mitigated

17% Global Traffic Growth and Starlink's 3.3x Surge

2024 Internet Traffic Trends | Cloudflare Radar

Global Traffic Growth

+17%

Year-over-Year
2024 vs 2023

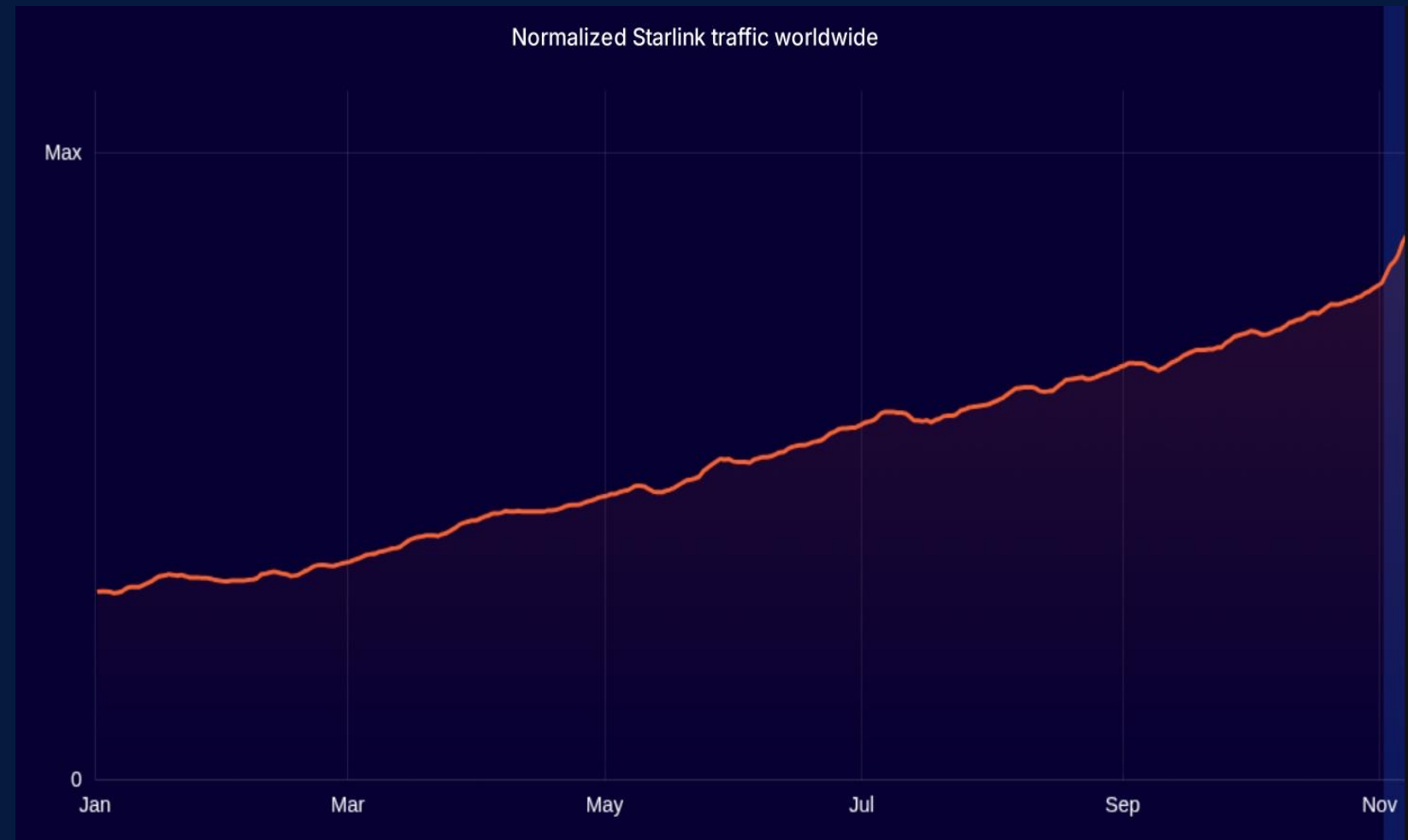
Starlink Traffic Surge

3.3x

SpaceX Starlink Growth
2023 → 2024

- Trend normalized from mid-Jan to remove post-holiday effects
- Starlink expanding satellite Internet to underserved regions globally
- Starlink not yet available worldwide; data scaled to peak request volume

Starlink Normalized Traffic — 2024



Google Leads Top 10 Services as Generative AI Rises

Ranking based on Cloudflare 1.1.1.1 DNS resolver aggregate data

Rank	Service	Description
#1	Google	Search, Cloud & Ads platform
#2	Facebook	Social media & messaging
#3	Apple	Devices, services & App Store
#4	TikTok	Short-form video platform
#5	AWS	Cloud infrastructure leader
#6	Microsoft	Enterprise software & cloud
#7	Instagram	Photo & video social network
#8	YouTube	Video streaming platform
#9	Amazon	E-commerce & marketplace
#10	WhatsApp	Encrypted messaging app

Generative AI continued its meteoric rise throughout 2024 — while the Metaverse receded. AI assistants, image generators, and code tools surged in DNS query volume, reflecting broad adoption across consumer and enterprise segments.

Top 10 Internet Services — 2024

Rank	Service
#1	Google
#2	Facebook
#3	Apple
#4	TikTok
#5	AWS
#6	Microsoft
#7	Instagram
#8	YouTube
#9	Amazon
#10	WhatsApp

Post-Quantum Encryption Jumps from 2% to 13% of TLS 1.3

Technology Adoption Shifts Observed in 2024

Post-Quantum Encryption Adoption — 2024



Chrome 124 enabled post-quantum key agreement by default on April 17, 2024 — triggering rapid adoption from ~2% to ~13% of TLS 1.3 traffic.

Post-Quantum Encryption

2% → 13%

of TLS 1.3 traffic in 2024

HTTP/3 Adoption

21%

of global web traffic

Top API Client Language

Go ~12%

of automated API requests

Mobile Traffic Share

41%

of global traffic from mobile

iOS Device Traffic

33%

of traffic from iOS devices

225 Major Disruptions: Government Shutdowns Lead Outage Causes

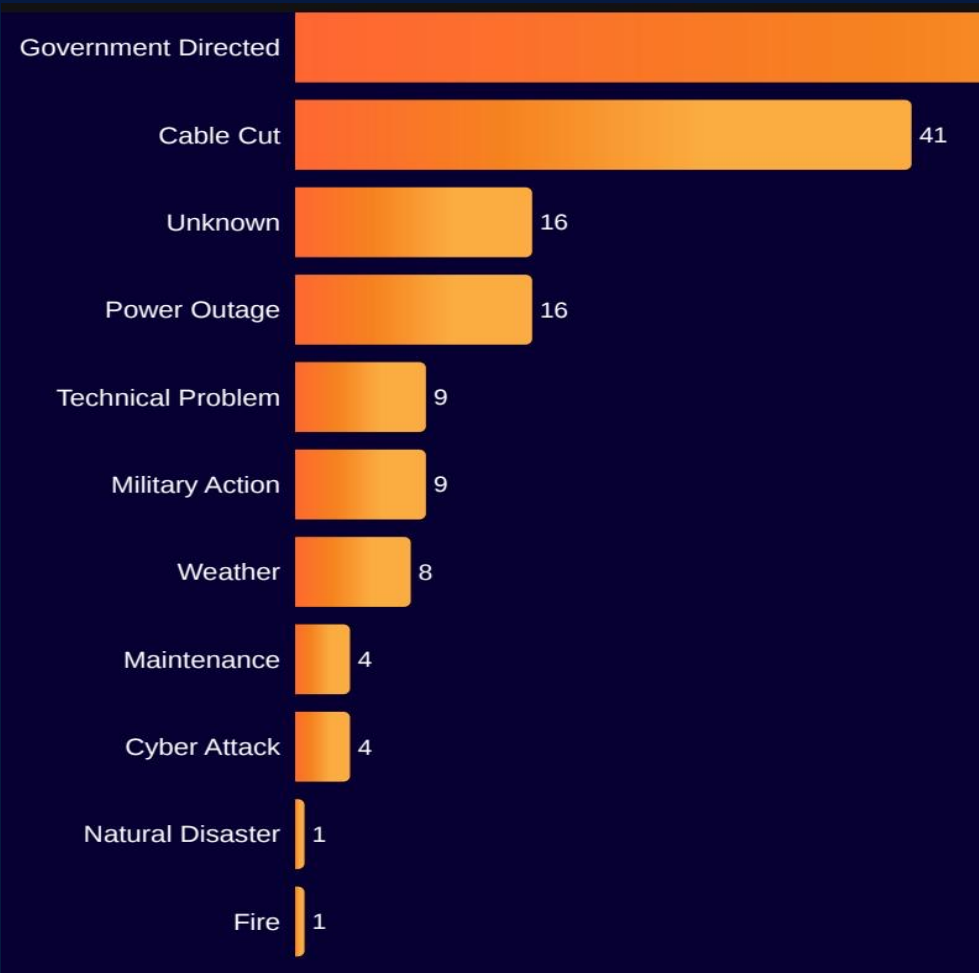
Cloudflare Radar Outage Center | Global Internet Disruptions 2024

225

Major Internet Disruptions
Observed in 2024

116+	Government-Directed Shutdowns
41	Submarine Cable Cuts
16	Power Outages
16	Unknown Cause
9	Technical Problems
9	Military Action

Disruption Causes — 2024



Cause	Count
Government Directed	116+
Cable Cut	41
Unknown	16
Power Outage	16
Technical Problem	9
Military Action	9
Weather	8
Maintenance	4
Cyber Attack	4
Natural Disaster	1
Fire	1

6.5% of Traffic Mitigated: Persistent Threats and Bot Concentration

Security Findings | Cloudflare Network | 2024

6.5%

Traffic Mitigated

4.3%

Emails Malicious

43%

Malicious Emails w/ Deceptive Link

68%

Bot Traffic from Top 10 Countries

6.4%

RPKI Valid Route Increase

Most Attacked Sectors

- 1st Gambling & Games — most targeted industry
- Bot and DDoS attacks dominate attack vectors
- 68% of global bot traffic from just top 10 countries
- Automated attack tooling increasingly accessible

Log4j Persistence (CVE-2021-44228)

- First disclosed December 2021 — still actively exploited
- Highlights danger of unpatched legacy vulnerabilities
- Attackers maintain persistent targeting 3+ years later
- RPKI valid routes +6.4% (routing security improvement)

Email Threat Metrics

- 4.3% of all emails processed were malicious
- 43% of malicious emails contained deceptive links
- .bar TLD originated most malicious & spam email
- Phishing via link deception remains dominant vector

Five Takeaways Shaping the 2025 Internet Outlook

Strategic Implications for Network and Security Leaders

01

Internet Traffic Growth Trajectory

Global traffic grew 17% YoY; Starlink's 3.3x surge signals satellite connectivity entering the mainstream. Capacity planning must account for accelerating demand from underserved regions.

02

Post-Quantum Encryption Momentum

Adoption leapt from 2% to 13% of TLS 1.3 traffic following Chrome 124's default enablement. Organizations should evaluate post-quantum readiness before broader browser and OS rollouts accelerate the curve further.

03

Connectivity Fragility from Government Actions

Government-directed shutdowns caused 116+ of 225 major disruptions — more than all technical causes combined. Resilience strategies must factor in geopolitical risk alongside infrastructure failure.

04

Persistent Security Threats at Scale

6.5% of all traffic required mitigation; Log4j exploitation continues 3+ years after disclosure. Patch velocity and threat-intelligence sharing remain critical — legacy vulnerabilities do not self-remediate.

05

Evolving Developer & Protocol Ecosystem

Go leads API client languages (~12%); HTTP/3 reached 21% adoption. The shift to modern protocols and languages reflects a developer ecosystem optimizing for performance, concurrency, and encrypted-by-default connectivity.